

Practical 4 - Estimating treatment effects using observational data

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `kidney_example.csv` and `kidney_example_continuousY.csv`

Introduction

This practical is in two parts. Part A uses the kidney stones example data from the Notes, in which the outcome Y is binary. Part B considers data on a similar study but with a continuous outcome Y . The other variables are the same: X denotes a binary treatment and Z a binary confounder. Both data sets include 700 individuals.

Variable	Description
y	Binary: 0 = success; 1 = failure Continuous: a score on a quality of life questionnaire
x	Treatment: 0 = Surgery; 1 = Lithotripsy
z	Size of kidney stone: 0 = small; 1 = large

Aims

The aims of this session are:

- To estimate conditional treatment effects for both binary and continuous outcomes.
- To estimate marginal treatment effects for both binary and continuous outcomes using standardization.
- To interpret estimates of marginal and conditional treatment effects.

Part A: Binary outcome

The data for this section are in the file `kidney_example.csv`. You can load the dataset into R using:

```
kidney <- read.csv("kidney_example.csv")
```

The data are summarised in the tables below.

	($X = 0$)	($X = 1$)	Total
Success ($Y = 0$)	273	289	562
Failure($Y = 1$)	77	61	138
Total	350	350	700

	Small stone size ($Z = 0$)		Large stone size ($Z = 1$)	
	Surgery ($X = 0$)	Lithotripsy ($X = 1$)	Surgery ($X = 0$)	Lithotripsy ($X = 1$)
Success ($Y = 0$)	81	234	192	55
Failure($Y = 1$)	6	36	71	25
Total	87	270	263	80

Question 1

In this question we will use logistic regression to obtain estimates of the conditional odds ratio(s)

$$\frac{\Pr(Y = 1|do(X = 1), Z = z) / \Pr(Y = 0|do(X = 1), Z = z)}{\Pr(Y = 1|do(X = 0), Z = z) / \Pr(Y = 0|do(X = 0), Z = z)}$$

for $z = 0, 1$.

- First using a logistic regression model for Y with X and Z as explanatory variables (not including their interaction).

SOLUTION

```
y_xz <- glm(y~x+z, data=kidney,  
            family="binomial")  
library(broom)
```

Warning: package 'broom' was built under R version 4.3.3

```
tidy(y_xz, exponentiate=TRUE,
     conf.int=TRUE)
```

A tibble: 3 x 7

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	(Intercept)	0.101	0.247	-9.29	1.56e-20	0.0613	0.161
2	x	1.43	0.229	1.56	1.19e- 1	0.913	2.24
3	z	3.53	0.239	5.27	1.33e- 7	2.22	5.68

The model being fitted is of the form

$$\Pr(Y = 1|X = x, Z = z) = \frac{e^{\beta_0 + \beta_x x + \beta_z z}}{1 + e^{\beta_0 + \beta_x x + \beta_z z}}.$$

The conditional OR is e^{β_x} , which is the same for $Z = 0, 1$ under this model. The estimate of this is $e^{\hat{\beta}_x} = 1.42$ (95% CI (0.91, 2.24)).

- ii) Using a logistic regression model for Y with X , Z and their interaction $X \times Z$ as explanatory variables.

SOLUTION

The model being fitted is of the form

$$\Pr(Y = 1|X = x, Z = z) = \frac{e^{\beta_0 + \beta_x x + \beta_z z + \beta_{xz} xz}}{1 + e^{\beta_0 + \beta_x x + \beta_z z + \beta_{xz} xz}}.$$

The conditional OR given $Z = 0$ is e^{β_x} and the conditional OR given $Z = 1$ is $e^{\beta_x + \beta_{xz}}$.

```
y_xz_int <- glm(y~x*z, data=kidney,
               family="binomial")
summary(y_xz_int)
```

Call:

```
glm(formula = y ~ x * z, family = "binomial", data = kidney)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.6027	0.4231	-6.152	7.67e-10 ***
x	0.7309	0.4594	1.591	0.111631
z	1.6079	0.4453	3.611	0.000305 ***

```
x:z          -0.5245      0.5372  -0.976  0.328821
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 694.98  on 699  degrees of freedom
Residual deviance: 661.86  on 696  degrees of freedom
AIC: 669.86
```

Number of Fisher Scoring iterations: 5

```
#log OR for effect of X in Z=0 is 0.7309, or programmatically:
coef(y_xz_int)['x']
```

```
      x
0.7308875
```

```
#log OR for effect of X in Z=1 is 0.7309-0.5245
coef(y_xz_int)['x'] + coef(y_xz_int)['x:z']
```

```
      x
0.2063581
```

The OR for $Z = 0$ and its 95% CI can be obtained using

```
exp(0.7309)
```

```
[1] 2.076949
```

```
#95% CI is
exp(0.7309-1.96*0.4594)
```

```
[1] 0.8440665
```

```
exp(0.7309+1.96*0.4594)
```

```
[1] 5.110637
```

To get a 95% CI for the OR of the X effect in the $Z = 1$ group, we can use the `comparisons` function in the `marginalEffects` package:

```
library(marginalEffects)
comparisons(y_xz_int,
            variables = "x",
            by = "z",
            type = "link",
            transform = "exp")
```

z	Estimate	Pr(> z)	S	2.5 %	97.5 %
0	2.08	0.112	3.2	0.844	5.11
1	1.23	0.458	1.1	0.712	2.12

Term: x
 Type: link
 Comparison: 1 - 0

Thus our estimates are $e^{\hat{\beta}_x} = 2.08$ (95% CI (0.84,5.11)) $e^{\hat{\beta}_x + \hat{\beta}_{xz}} = 1.23$ (95% CI (0.71,2.12))

Question 2

Using your model from Question 1(ii) obtain estimates of:

- i) The conditional probabilities $\Pr(Y = 1|do(X = 1), Z = z)$ and $\Pr(Y = 1|do(X = 0), Z = z)$ for $z = 0, 1$. One approach is to use the `fitted.values` in your fitted model object.

SOLUTION

```
summary(y_xz_int$fitted.values[kidney$x==0 & kidney$z==0])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.06897	0.06897	0.06897	0.06897	0.06897	0.06897

```
summary(y_xz_int$fitted.values[kidney$x==0 & kidney$z==1])
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.27	0.27	0.27	0.27	0.27	0.27

```
summary(y_xz_int$fitted.values[kidney$x==1 & kidney$z==0])
```

```
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.1333  0.1333  0.1333  0.1333  0.1333  0.1333
```

```
summary(y_xz_int$fitted.values[kidney$x==1 & kidney$z==1])
```

```
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.3125  0.3125  0.3125  0.3125  0.3125  0.3125
```

```
#another option: using predict
predict(y_xz_int,newdata=data.frame(x=c(0,0,1,1),z=c(0,1,0,1)),type="response")
```

```
      1      2      3      4
0.06896552 0.26996198 0.13333333 0.31250000
```

ii) The conditional risk differences

$$\Pr(Y = 1|do(X = 1), Z = z) - \Pr(Y = 1|do(X = 0), Z = z), \quad z = 0, 1.$$

SOLUTION The estimates of the conditional risk differences are:

$$\widehat{\Pr}(Y = 1|do(X = 1), Z = 0) - \widehat{\Pr}(Y = 1|do(X = 0), Z = 0) = 0.1333 - 0.0687 = 0.065$$

$$\widehat{\Pr}(Y = 1|do(X = 1), Z = 1) - \widehat{\Pr}(Y = 1|do(X = 0), Z = 1) = 0.3125 - 0.27 = 0.043$$

iii) The conditional risk ratios

$$\frac{\Pr(Y = 1|do(X = 1), Z = z)}{\Pr(Y = 1|do(X = 0), Z = z)}, \quad z = 0, 1.$$

SOLUTION

The estimates of the conditional risk ratios are:

$$\widehat{\Pr}(Y = 1|do(X = 1), Z = 0) / \widehat{\Pr}(Y = 1|do(X = 0), Z = 0) = \frac{0.1333}{0.06897} = 1.93$$

$$\widehat{\Pr}(Y = 1|do(X = 1), Z = 1) / \widehat{\Pr}(Y = 1|do(X = 0), Z = 1) = \frac{0.3125}{0.27} = 1.16$$

Discuss: Compare your results from the logistic regression models with those given in Table 4.2 in the Notes (which were not obtained using regression). Interpret your conditional treatment effect estimates obtained in questions 1 and 2. What assumption do we make to interpret these estimates ‘causally’?

SOLUTION

The results from the logistic regression model including the interaction between X and Z give the same conditional OR estimates to those seen in the notes (except for some rounding errors in the results given in the notes!). The interpretations of the conditional estimates are as follows.

Conditional ORs: In those with small stone size ($Z = 0$) the odds of a bad outcome ($Y = 1$) under lithotripsy are estimated to be 2.08 times the odds of a bad outcome under surgery. In those with large stone size ($Z = 1$) the odds of a bad outcome ($Y = 1$) under lithotripsy are estimated to be 1.23 times the odds of a bad outcome under surgery. However, the 95% CIs for both estimates include 1. There is in fact no evidence that Z modifies the effect of treatment on outcome, as the OR for the interaction term is 0.59, with 95 CI (0.21,1.70), and corresponding p-value 0.329.

Conditional risk differences: In those with small stone size ($Z = 0$) the absolute risk of a bad outcome ($Y = 1$) under lithotripsy is estimated to 0.064 higher than the risk under surgery. In those with large stone size ($Z = 1$) the absolute risk of a bad outcome ($Y = 1$) under lithotripsy is estimated to 0.043 higher than the risk under surgery.

Conditional risk ratios: In those with small stone size ($Z = 0$) the risk of a bad outcome ($Y = 1$) under lithotripsy is estimated to be 1.93 times the risk of a bad outcome under surgery. In those with large stone size ($Z = 1$) the risk of a bad outcome ($Y = 1$) under lithotripsy are estimated to be 1.16 times the risk of a bad outcome under surgery.

Question 3

We will next estimate marginal treatment effects.

- i) Using the results from your model in Question 1(ii), and information about $\Pr(Z = z)$, use the standardization formula in (4.8) to estimate $\Pr(Y = 1|do(X = 1))$ and $\Pr(Y = 1|do(X = 0))$. The standardization formula is:

$$\Pr(Y = 1|do(X = x)) = \sum_{z=0,1} \Pr(Y = 1|X = x, Z = z) \Pr(Z = z)$$

Note that the empirical/estimated probabilities of having small and large kidney stone size are $\widehat{\Pr}(Z = 0) = 0.51$ and $\widehat{\Pr}(Z = 1) = 0.49$ (check you agree!).

- ii) Using your results from (i) obtain estimates of the marginal risk difference, marginal risk ratio, and marginal odds ratio, as defined in equations (4.1)-(4.3) in the Notes.

You may find it useful to use the table below to keep track of your estimates.

$\Pr(Y = 1 do(X = 1))$	$\Pr(Y = 1 X = 1, Z = 0)$	
	$\Pr(Y = 1 X = 1, Z = 1)$	
	$\Pr(Y = 1 X = 1, Z = 0) \times \Pr(Z = 0)$	
	$\Pr(Y = 1 X = 1, Z = 1) \times \Pr(Z = 1)$	
	$\Pr(Y = 1 X = 1, Z = 0) \times \Pr(Z = 0)$ $+ \Pr(Y = 1 X = 1, Z = 1) \times \Pr(Z = 1)$	
$\Pr(Y = 1 do(X = 0))$	$\Pr(Y = 1 X = 0, Z = 0)$	
	$\Pr(Y = 1 X = 0, Z = 1)$	
	$\Pr(Y = 1 X = 0, Z = 0) \times \Pr(Z = 0)$	
	$\Pr(Y = 1 X = 0, Z = 1) \times \Pr(Z = 1)$	
	$\Pr(Y = 1 X = 0, Z = 0) \times \Pr(Z = 0)$ $+ \Pr(Y = 1 X = 0, Z = 1) \times \Pr(Z = 1)$	

SOLUTION

See estimates below.

$\Pr(Y = 1 do(X = 1))$	$\Pr(Y = 1 X = 1, Z = 0)$	0.133
	$\Pr(Y = 1 X = 1, Z = 1)$	0.313
	$\Pr(Y = 1 X = 1, Z = 0) \times \Pr(Z = 0)$	0.068
	$\Pr(Y = 1 X = 1, Z = 1) \times \Pr(Z = 1)$	0.153
	$\Pr(Y = 1 X = 1, Z = 0) \times \Pr(Z = 0)$ $+ \Pr(Y = 1 X = 1, Z = 1) \times \Pr(Z = 1)$	0.221
$\Pr(Y = 1 do(X = 0))$	$\Pr(Y = 1 X = 0, Z = 0)$	0.069
	$\Pr(Y = 1 X = 0, Z = 1)$	0.270
	$\Pr(Y = 1 X = 0, Z = 0) \times \Pr(Z = 0)$	0.035
	$\Pr(Y = 1 X = 0, Z = 1) \times \Pr(Z = 1)$	0.132
	$\Pr(Y = 1 X = 0, Z = 0) \times \Pr(Z = 0)$ $+ \Pr(Y = 1 X = 0, Z = 1) \times \Pr(Z = 1)$	0.167

The marginal risk difference estimate is:

$$\widehat{\Pr}(Y = 1|do(X = 1)) - \widehat{\Pr}(Y = 1|do(X = 0)) = 0.054$$

The marginal risk ratio estimate is:

$$\frac{\widehat{\Pr}(Y=1|do(X=1))}{\widehat{\Pr}(Y=1|do(X=0))} = 1.32$$

The marginal odds ratio estimate is:

$$\frac{\widehat{\Pr}(Y=1|do(X=1))/\widehat{\Pr}(Y=0|do(X=1))}{\widehat{\Pr}(Y=1|do(X=0))/\widehat{\Pr}(Y=0|do(X=0))} = 1.41$$

Discuss: What are the interpretations of the causal marginal treatment effects you have estimated? Compare these with the conditional treatment effects estimated in questions 1 and 2.

SOLUTION

Marginal risk difference: If everyone had received lithotripsy we would expect the risk of a bad outcome to be 0.054 higher than if everyone had received surgery.

Marginal risk ratio: If everyone had received lithotripsy we would expect the risk of a bad outcome to be 1.32 times that (i.e. 32% higher) if everyone had received surgery.

Marginal odds ratio: If everyone had received lithotripsy we would expect the odds of a bad outcome to be 1.41 times that (i.e. 41% higher) if everyone had received surgery.

Question 4

The `marginalEffects` package can also be used to estimate the risk difference more directly by standardization.

- i) The `avg_predictions` function can be used to obtain estimates of $\Pr(Y = 1|do(X = 0))$ and $\Pr(Y = 1|do(X = 1))$ using the same standardization procedure that you used above:

```
avg_predictions(y_xz_int, variables = "x")
```

SOLUTION

```
avg_predictions(y_xz_int,  
                variables = "x")
```

x	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0	0.167	0.0193	8.68	<0.001	57.9	0.130	0.205
1	0.221	0.0275	8.04	<0.001	50.0	0.167	0.275

Type: response

- ii) Estimates of the marginal risk difference can be obtained using:

```
avg_comparisons(y_xz_int, variables = "x")
```

Run this. What do you conclude about the effect of lithotripsy compared to surgery on the chance of failure?

SOLUTION

```
avg_comparisons(y_xz_int,  
                variables = "x")
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0.0537	0.0336	1.6	0.11	3.2	-0.0122	0.119

Term: x
Type: response
Comparison: 1 - 0

The marginal risk difference estimate is the same as the one obtained earlier by hand - the risk of failure were the population to all be given lithotripsy is estimated to be 0.054 higher than if all were given surgery. The confidence interval however includes zero and the p-value is not statistically significant ($p = 0.11$), so there is no evidence that in fact the population risk would differ depending on which intervention was given.

Question 5

Lastly we will use the empirical standardization method to estimate the marginal probabilities $\Pr(Y = 1|do(X = 0))$ and $\Pr(Y = 1|do(X = 1))$. This is the method that `avg_predictions` is using, but breaking down the steps involved should help consolidate your understanding of the method. Run the following code, making sure you understand what is happening at each step. Check that you get the same estimates as we obtained previously.

```
pred_covs <- subset(kidney,select=c(x,z))
#set treatment to 0
pred_covs$x <- 0
#manually calculate predicted probabilities for each individual
#use plogis(x)=exp(x)/(1+exp(x))
pred_covs$pred0 = with(pred_covs,
  plogis(-2.60269+0.7308873*x+1.607874*z-0.5245292*x*z))
#now set treatment to 1
pred_covs$x <- 1
pred_covs$pred1 = with(pred_covs,
  plogis(-2.60269+0.7308873*x+1.607874*z-0.5245292*x*z))

#calculate the mean predicted outcome in each scenario
mean(pred_covs$pred0)
```

```
[1] 0.1674537
```

```
mean(pred_covs$pred1)
```

```
[1] 0.2211249
```

```
#risk difference is  
mean(pred_covs$pred1)-mean(pred_covs$pred0)
```

```
[1] 0.05367119
```

Part B: Continuous outcome

In the second part of this practical we will use a modified version of the kidney study data with a continuous outcome Y instead of a binary outcome. For example, Y could be a score on a quality of life questionnaire. The data are available in the file `kidney_example_continuousY.csv`. Load the data.

Question 6

What are the means of Y in each of the four groups defined by X and Z ? That is, what are $E(Y|X = x, Z = z)$ for $x, z = 0, 1$?

SOLUTION

```
kidney_cts <- read.csv("kidney_example_continuousY.csv")  
mean(kidney_cts$y[(kidney_cts$x==0) & (kidney_cts$z==0)])
```

```
[1] 0.2012412
```

```
mean(kidney_cts$y[(kidney_cts$x==0) & (kidney_cts$z==1)])
```

```
[1] -0.8482609
```

```
mean(kidney_cts$y[(kidney_cts$x==1) & (kidney_cts$z==0)])
```

```
[1] 0.556622
```

```
mean(kidney_cts$y[(kidney_cts$x==1) & (kidney_cts$z==1)])
```

```
[1] 0.1299626
```

Question 7

Fit a linear regression of Y on X , Z and their interaction $X \times Z$. How do the coefficients relate to the means of Y in each of the four groups defined by X and Z ?

SOLUTION

```
ycts_xz_int <- lm(y~x*z, data=kidney_cts)
summary(ycts_xz_int)
```

Call:

```
lm(formula = y ~ x * z, data = kidney_cts)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-11.6238	-2.0991	0.0091	2.0363	8.6253

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.2012	0.3236	0.622	0.53427
x	0.3554	0.3721	0.955	0.33993
z	-1.0495	0.3733	-2.811	0.00508 **
x:z	0.6228	0.5358	1.163	0.24541

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.019 on 696 degrees of freedom

Multiple R-squared: 0.04154, Adjusted R-squared: 0.03741

F-statistic: 10.06 on 3 and 696 DF, p-value: 1.714e-06

The linear regression model is of the form:

$$E(Y|X = x, Z = z) = \beta_0 + \beta_X x + \beta_Z z + \beta_{XZ} xz.$$

The above expectations relate to the model parameters as follows:
 $E(Y|X = 0, Z = 0) = \beta_0$
 $E(Y|X = 0, Z = 1) = \beta_0 + \beta_Z$
 $E(Y|X = 1, Z = 0) = \beta_0 + \beta_X$
 $E(Y|X = 1, Z = 1) = \beta_0 + \beta_X + \beta_Z + \beta_{XZ}$

Question 8

Use your results from question 6 or 7 to estimate the conditional effect of X on Y given Z for $Z = 0, 1$, using a conditional mean difference:

$$\begin{aligned} &E(Y|do(X = 1), Z = 0) - E(Y|do(X = 0), Z = 0) \\ &E(Y|do(X = 1), Z = 1) - E(Y|do(X = 0), Z = 1) \end{aligned}$$

Discuss: interpret the estimates. What assumption do you need to make in order to interpret the conditional mean differences as causal treatment effect estimates?

SOLUTION

The conditional mean difference estimates are

$$\begin{aligned} &\widehat{E}(Y|do(X = 1), Z = 0) - \widehat{E}(Y|do(X = 0), Z = 0) \\ &= \widehat{E}(Y|X = 1, Z = 0) - \widehat{E}(Y|X = 0, Z = 0) \\ &= \hat{\beta}_X = 0.355 \end{aligned}$$

and

$$\begin{aligned} &\widehat{E}(Y|do(X = 1), Z = 1) - \widehat{E}(Y|do(X = 0), Z = 1) \\ &= \widehat{E}(Y|X = 1, Z = 1) - \widehat{E}(Y|X = 0, Z = 1) \\ &= \hat{\beta}_X + \hat{\beta}_{XZ} = 0.978 \end{aligned}$$

The confidence interval for $\hat{\beta}_X$ can be obtained by

```
0.3554-qt(0.975,df=696)*0.3721
```

```
[1] -0.375173
```

```
0.3554+qt(0.975,df=696)*0.3721
```

```
[1] 1.085973
```

Since the residual degrees of freedom is large (696), we get essentially the same answer with:

```
0.3554-1.96*0.3721
```

```
[1] -0.373916
```

```
0.3554+1.96*0.3721
```

```
[1] 1.084716
```

For $\hat{\beta}_X + \hat{\beta}_{XZ}$ we can use `comparisons` in the `marginaleffects` package to handle the calculation of the standard error, which as we have discussed previously, needs to incorporate both the uncertainty in the two parameter estimates but also their covariance. To do this we specify the `newdata` argument as `datagrid(z=1)` to specify we want the contrast performed holding $Z = 1$, and `variables="x"` to specify we want the effect of X .

```
comparisons(  
  ycts_xz_int,  
  variables = "x",  
  newdata = datagrid(z = 1),  
  df= "residual"  
)
```

	z	Estimate	Std. Error	t	Pr(> t)	S	2.5 %	97.5 %	Df
1	0.978	0.385	2.54	0.0114	6.5	0.221	1.73	696	

Term: x

Type: response

Comparison: 1 - 0

In those with $Z = 0$, if everyone received lithotripsy the expected outcome would be 0.355 higher than if everyone had received surgery. In those with $Z = 1$, if everyone received lithotripsy the expected outcome would be 0.978 higher than if everyone had received surgery.

Question 9

We now estimate the causal marginal treatment effect.

- i) By hand, use standardization to complete the table below, and hence to estimate the causal marginal treatment effect:

$$E(Y|do(X = 1)) - E(Y|do(X = 0))$$

Reminder, by standardization $E(Y|do(X = x))$ is given by

$$E(Y|X = x, Z = 0) \times \Pr(Z = 0) + E(Y|X = x, Z = 1) \times \Pr(Z = 1)$$

$E(Y do(X = 1))$	$E(Y X = 1, Z = 0)$	
	$\Pr(Z = 0)$	0.51
	$E(Y X = 1, Z = 1)$	
	$\Pr(Z = 1)$	0.49
	$E(Y do(X = 1))$	
$E(Y do(X = 0))$	$E(Y X = 0, Z = 0)$	
	$\Pr(Z = 0)$	0.51
	$E(Y X = 0, Z = 1)$	
	$\Pr(Z = 1)$	0.49
	$E(Y do(X = 0))$	
	$E(Y do(X = 1)) - E(Y do(X = 0))$	

SOLUTION

See table below.

$E(Y do(X = 1))$	$E(Y X = 1, Z = 0)$	0.557
	$\Pr(Z = 0)$	0.51
	$E(Y X = 1, Z = 1)$	0.130
	$\Pr(Z = 1)$	0.49
	$E(Y do(X = 1))$	0.347
$E(Y do(X = 0))$	$E(Y X = 0, Z = 0)$	0.201
	$\Pr(Z = 0)$	0.51
	$E(Y X = 0, Z = 1)$	-0.848
	$\Pr(Z = 1)$	0.49
	$E(Y do(X = 0))$	-0.313
	$E(Y do(X = 1)) - E(Y do(X = 0))$	0.661

ii) You can also, as before, obtain the same estimate using the `marginalEffects` package:

```
#predicted mean outcome under setting x to 0 or 1
avg_predictions(ycts_xz_int,
  variables = "x",
  df="residual")
```

x	Estimate	Std. Error	t	Pr(> t)	S	2.5 %	97.5 %	Df
0	-0.313	0.189	-1.66	0.0974	3.4	-0.6833	0.0572	696
1	0.348	0.190	1.83	0.0679	3.9	-0.0256	0.7207	696

Type: response

```
#and their difference
avg_comparisons(ycts_xz_int,
  variables = "x",
  df="residual")
```

```
Estimate Std. Error    t Pr(>|t|)    S 2.5 % 97.5 % Df
      0.661      0.268 2.47  0.0139 6.2 0.135   1.19 696
```

```
Term: x
Type: response
Comparison: 1 - 0
```

Question 10

Modify the code given in question 5 to estimate the marginal mean difference treatment effect using empirical standardisation.

SOLUTION

```
pred_covs <- subset(kidney,select=c(x,z))
#set treatment to 0
pred_covs$x <- 0
#manually calculate predicted outcomes
pred_covs$pred0 = with(pred_covs,
  0.2012412+0.3553808*x-1.049502*z+0.6228426*x*z)
#not set treatment to 1
pred_covs$x <- 1
pred_covs$pred1 = with(pred_covs,
  0.2012412+0.3553808*x-1.049502*z+0.6228426*x*z)

#calculate the mean predicted outcome in each scenario
mean(pred_covs$pred0)
```

```
[1] -0.3130148
```

```
mean(pred_covs$pred1)
```

```
[1] 0.3475589
```

```
#causal mean difference is
mean(pred_covs$pred1)-mean(pred_covs$pred0)
```

```
[1] 0.6605737
```

Question 11

Fit a linear regression of Y on X and Z, without an interaction term between X and Z, storing it in an object called `ycts_xz_noint`. What is the estimated conditional mean difference $E(Y|do(X = 1), Z = z) - E(Y|do(X = 0), Z = z)$ for $Z = 0, 1$? Compare this with the result from using:

```
avg_comparisons(ycts_xz_noint,
                 variables = "x")
```

SOLUTION

```
ycts_xz_noint <- lm(y~x+z,data=kidney_cts)
summary(ycts_xz_noint)
```

Call:

```
lm(formula = y ~ x + z, data = kidney_cts)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-11.6990	-2.1232	0.0001	1.9913	8.8525

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.02603	0.25798	-0.101	0.91966
x	0.65588	0.26778	2.449	0.01456 *
z	-0.74705	0.26784	-2.789	0.00543 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.019 on 697 degrees of freedom

Multiple R-squared: 0.03968, Adjusted R-squared: 0.03693

F-statistic: 14.4 on 2 and 697 DF, p-value: 7.443e-07

Fitting a linear regression of Y on X and Z without the interaction gives an estimated conditional mean difference of 0.656, with 95% CI (0.130,1.182). This is the expected mean difference in both the $Z = 0$ and $Z = 1$ groups, under the assumption of no interaction between X and Z . The interpretation is that if everyone in the $Z = 0$ group had received lithotripsy we would expect the mean outcome to be 0.656 higher than had everyone in the $Z = 0$ group received surgery. Similarly, if everyone in the $Z = 1$ group had received lithotripsy we would expect the mean outcome to be 0.656 higher than had everyone in the $Z = 1$ group received surgery. It was not asked for in the question, but in fact there is no evidence that Z modifies the effect of X on Y .

Using `avg_comparisons`, we obtain:

```
avg_comparisons(ycts_xz_noint,
                variables = "x")
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0.656	0.268	2.45	0.0143	6.1	0.131	1.18

Term: x

Type: response

Comparison: 1 - 0

the same estimated marginal mean difference of 0.656. The interpretation is that if everyone had received lithotripsy we would expect the mean outcome to be 0.656 higher than had everyone received surgery. If we assume no interaction in the model, the conditional and marginal mean difference effects are equal (by assumption).